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AI for Bharat Hackathon

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Team Name : DEVOPS

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Problem Statement: An AI- and NLP-based system that automatically analyzes Terms & Conditions documents to extract key clauses, assess risks, generate simple summaries, and deliver voice-based explanations to enhance user understanding and decision-making.

AI Guardian

For Terms & Conditions

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Most people accept lengthy, complex Terms & Conditions without reading them — leading to hidden fees, surprise auto-renewals, and data misuse. **Our product is an AI-powered digital guardian that protects you before you click "Agree".**

The Problem

Lengthy legal text, low legal literacy, and time pressure cause **unconscious consent**. Users unknowingly agree to costly consequences and forfeited rights.

The Solution

An AI app that analyzes T&Cs, flags risky clauses, explains them in **plain English**, and produces a clear 0–100 Risk Score for instant decision making.

Target Users

OTT subscribers, banking & fintech customers, insurance buyers, students, and **digitally vulnerable users** entering the online ecosystem.

HOW IT WORKS?

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THE WORKFLOW

1. Ingest

User uploads PDF/text or pastes site content. System tokenizes & normalizes legal language.

2. Analyze

Advanced NLU models detect obligations, permissions, auto-renewals, & risky constructs.

3. Score & Explain

0-100 Risk Score computed. High-risk clauses highlighted with plain-English explanations.

4. Alert Before Consent

Real-time alerts & concise warnings appear before consent, reducing costly mistakes.

KEY DIFFERENTIATORS

Risk Detection

Clause-level identification of dark patterns like auto-renewals, hidden fees, and forced arbitration.

Clear Risk Score

Numeric score (**0–100**) for instant, comparable judgement across services.

Plain English

Simple explanations and recommended actions designed specifically for non-technical users.

Market Fit (India Ready)

Tailored for Indian fintech, OTT, and banking specifics with localization and regulatory awareness.

> Compared to passive summarizers, our platform proactively quantifies risk, detects dark patterns, and delivers actionable alerts at the moment of consent — turning opaque legalese into clear, timely consumer protection.

TERMS & CONDITIONS ANALYSIS PROCESS

One-Click T&C Summary

Instantly converts long, complex legal documents into simple, digestible bullet points.

Color-Coded Clause Highlighting

Automatically categorizes and highlights clauses based on their potential risk level.

"One-Line Truth" Feature

Summarizes the entire agreement's core meaning into a single, truthful statement.

Browser & App Integration

Seamlessly integrates with web browsers and apps for real-time analysis.

Risk & Warning Detection

Proactively identifies and alerts users to potential legal risks and hidden warnings.

Risk Score Indicator

Calculates and provides an overall numerical risk score (0-100) for quick assessment.

Multilingual & Voice Support

Offers full translation capabilities and audio options for accessibility.

AI-Powered T&C Analysis Workflow

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1. **T&C Submission**
   User uploads or pastes Terms & Conditions
2. **Document Parsing**
   AI extracts text from the document structure
3. **NLP Analysis**
   AI analyzes text for meaning and semantics
4. **Clause Detection**
   Identifies individual clauses within text
5. **Risk Assessment**
   Classifies clauses as high, medium, or safe
6. **Summary Creation**
   Generates a concise, readable summary
7. **Risk Scoring**
   Calculates overall risk score for the T&Cs
8. **Dashboard Display**
   Presents summary & scores on dashboard

Solution Architecture

AT A GLANCE

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Frontend
React, Flutter

Backend
Node.js, FastAPI

Document Processing
PyPDF2, Tesseract OCR

NLP & ML
spaCy, NLTK, Transformers, Scikit-learn, TensorFlow

Data Storage
MongoDB, Firebase

Voice
GoogleTTS, Whisper AI

Visualization
Chart.js, Recharts

Browser Extension
JavaScript, Chrome API

End-to-End Process Flow

SEAMLESS INTEGRATION FROM INPUT TO INSIGHT

User Input
Document upload & initial query submission via interface

T&C Processing
OCR extraction & intelligent text parsing

NLP Risk Engine
AI analysis to identify hidden risks & clauses

Visual & Voice
Generating risk visualizations & audio summaries

Extension
Real-time alerts via browser integration

KEY TAKEAWAY

Each stage of this process flow seamlessly connects to deliver a comprehensive solution. From intelligent **risk assessment** to **direct browser interaction**, the system ensures users receive immediate, actionable insights in the format that suits them best.

PROPOSED SOLUTION - PROTOTYPE

Client Extension

Detects TAC URL & page content triggers.

REACT / CHROME API

Text Extractor

Dynamic DOM scraping & boilerplate removal.

PUPPETEER / CHEERIO

AWS CLOUD INFRASTRUCTURE

Security Layer

PII Anonymization & Data Scrubbing.

AWS LAMBDA

AI Summary Core

Categorized legal risk analysis.

AMAZON BEDROCK

Knowledge Base

Caching common summaries for speed.

AMAZON DYNAMODB

API Gateway

Scalable request routing & auth.

AMAZON API GATEWAY

Smart UI Overlay

Interactive "TL;DR" drawer on the web page.

SHADOW DOM

Risk Alerts

Highlighting hidden 'Red Flag' clauses.

CUSTOM ANALYTICS

1. Intelligent Trigger

System scans for keywords like 'Terms', 'Privacy', or 'EULA', ensuring the summary is ready before the user even scrolls.

2. Foundation Models

Leveraging **Claude 3** on **Amazon Bedrock** to process long-form legal text (up to 200k tokens) into simplified insights.

3. Privacy First

User identity is never sent to the LLM. All data is processed in a transient state within a VPC, ensuring absolute data

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Thank You